# Problem 1: Uncertainty Analysis of Cylindricity Measurement

Detailed analytical solution by the GUM method

## 1 Problem statement

The cylindricity error of a machined cylindrical component is measured using a cylindricity tester. Five repeated measurements are made, with repositioning and remeasurement of the component between trials:

| Measurement number | Measured cylindricity ($\mu$m) |
|---|---|
| 1 | 8.2 |
| 2 | 8.5 |
| 3 | 8.1 |
| 4 | 8.4 |
| 5 | 8.3 |

The measurement model is

$$C = C_{\text{ind}} + \delta_{\text{cal}} + \delta_{\text{res}} + \delta_{\text{align}},$$

where $C$ is the corrected cylindricity error ($\mu$m), $C_{\text{ind}}$ is the mean of the five indicated values, $\delta_{\text{cal}}$ is the calibration correction, $\delta_{\text{res}}$ is the resolution correction and $\delta_{\text{align}}$ is the setup and alignment correction.

| Input | Type | PDF | Parameters |
|---|---|---|---|
| Calibration $\delta_{\text{cal}}$ | B | Normal | correction $= 0$ $\mu$m, $U_{\text{cal}} = 0.6$ $\mu$m, $k = 2$ |
| Resolution $\delta_{\text{res}}$ | B | Uniform | resolution 0.1 $\mu$m, bounded by $\pm 0.05$ $\mu$m |
| Alignment $\delta_{\text{align}}$ | B | Triangular | centred at 0, limits $\pm 0.30$ $\mu$m |

Required: the uncertainty budget (mean, standard uncertainties, sensitivity coefficients, combined standard uncertainty) and the expanded uncertainty at a 95 % confidence level.

## 2 Standard uncertainty of the assumed distributions

The standard uncertainty of an input is the standard deviation of its probability density function $g(x)$:

$$u^2 = \int_{-\infty}^{\infty} (x - \mu)^2 \, g(x) \, dx.$$

### Rectangular (uniform) distribution on $[-a, a]$

Here $g(x) = \dfrac{1}{2a}$ for $-a \le x \le a$, and $\mu = 0$:

$$u^2 = \int_{-a}^{a} x^2 \frac{1}{2a} \, dx = \frac{1}{2a} \left[ \frac{x^3}{3} \right]_{-a}^{a} = \frac{1}{2a} \cdot \frac{2a^3}{3} = \frac{a^2}{3}, \qquad u = \frac{a}{\sqrt{3}}.$$

### Symmetric triangular distribution on $[-a, a]$

Here $g(x) = \dfrac{a - |x|}{a^2}$ for $-a \le x \le a$, and $\mu = 0$. Using symmetry,

$$u^2 = 2\int_0^a x^2\,\frac{a-x}{a^2}\,dx = \frac{2}{a^2}\left[\frac{ax^3}{3} - \frac{x^4}{4}\right]_0^a = \frac{2}{a^2}\left(\frac{a^4}{3} - \frac{a^4}{4}\right) = \frac{2}{a^2}\cdot\frac{a^4}{12} = \frac{a^2}{6}, \qquad u = \frac{a}{\sqrt{6}}.$$

### Normal distribution

If an expanded uncertainty $U$ is quoted with coverage factor $k$, the standard uncertainty is $u = U/k$.

## 3 Step 1: Best estimate of the measurand

All three corrections have an expected value of zero:

$$\delta_{\text{cal}} = \delta_{\text{res}} = \delta_{\text{align}} = 0 \quad \Longrightarrow \quad C = C_{\text{ind}}.$$

## 4 Step 2: Type A evaluation (repeatability)

### Mean

$$\bar{C}_{\text{ind}} = \frac{1}{n}\sum_{j=1}^{n} C_j = \frac{8.2 + 8.5 + 8.1 + 8.4 + 8.3}{5} = \frac{41.5}{5} = 8.30\ \mu\text{m}.$$

### Deviations from the mean

| $j$ | $C_j$ ($\mu$m) | $C_j - \bar{C}$ ($\mu$m) | $(C_j - \bar{C})^2$ ($\mu$m$^2$) |
|---|---|---|---|
| 1 | 8.2 | $-0.1$ | 0.01 |
| 2 | 8.5 | $+0.2$ | 0.04 |
| 3 | 8.1 | $-0.2$ | 0.04 |
| 4 | 8.4 | $+0.1$ | 0.01 |
| 5 | 8.3 | 0.0 | 0.00 |
| Sum | 41.5 | 0.0 | 0.10 |

### Experimental standard deviation

$$s^2 = \frac{1}{n-1}\sum_{j=1}^{n}(C_j - \bar{C})^2 = \frac{0.10}{5-1} = 0.025\ \mu\text{m}^2, \qquad s = \sqrt{0.025} = 0.15811\ \mu\text{m}.$$

### Standard uncertainty of the mean

$$u(C_{\text{ind}}) = \frac{s}{\sqrt{n}} = \frac{0.15811}{\sqrt{5}} = \frac{0.15811}{2.23607} = 0.07071\ \mu\text{m},$$

with degrees of freedom

$$\nu_A = n - 1 = 4.$$

## 5 Step 3: Type B evaluations

### Calibration (normal distribution)

The calibration certificate gives $U_{\text{cal}} = 0.6$ $\mu$m with $k = 2$:

$$u(\delta_{\text{cal}}) = \frac{U_{\text{cal}}}{k} = \frac{0.6}{2} = 0.30000 \ \mu\text{m}.$$

### Resolution (rectangular distribution)

A resolution of 0.1 $\mu$m gives a quantisation error bounded by $\pm 0.05$ $\mu$m, so $a = 0.05$ $\mu$m:

$$u(\delta_{\text{res}}) = \frac{a}{\sqrt{3}} = \frac{0.05}{1.73205} = 0.02887 \ \mu\text{m}.$$

### Setup and alignment (triangular distribution)

The limits are $\pm 0.30$ $\mu$m, so $a = 0.30$ $\mu$m:

$$u(\delta_{\text{align}}) = \frac{a}{\sqrt{6}} = \frac{0.30}{2.44949} = 0.12247 \ \mu\text{m}.$$

All three Type B evaluations have $\nu = \infty$.

## 6 Step 4: Sensitivity coefficients

The sensitivity coefficient of each input is the partial derivative of the model with respect to that input:

$$c_{\text{ind}} = \frac{\partial C}{\partial C_{\text{ind}}} = 1, \qquad c_{\text{cal}} = \frac{\partial C}{\partial \delta_{\text{cal}}} = 1, \qquad c_{\text{res}} = \frac{\partial C}{\partial \delta_{\text{res}}} = 1, \qquad c_{\text{align}} = \frac{\partial C}{\partial \delta_{\text{align}}} = 1.$$

Because the model is a simple sum, every input enters with unit weight.

## 7 Step 5: Combined standard uncertainty

For uncorrelated inputs, the law of propagation of uncertainty gives

$$u_c^2(C) = \sum_i c_i^2 \, u^2(x_i) = c_{\text{ind}}^2 u^2(C_{\text{ind}}) + c_{\text{cal}}^2 u^2(\delta_{\text{cal}}) + c_{\text{res}}^2 u^2(\delta_{\text{res}}) + c_{\text{align}}^2 u^2(\delta_{\text{align}}).$$

Substituting the values:

$$\begin{aligned} u_c^2(C) &= (1 \times 0.07071)^2 + (1 \times 0.30000)^2 + (1 \times 0.02887)^2 + (1 \times 0.12247)^2 \\ &= 0.005000 + 0.090000 + 0.000833 + 0.015000 \\ &= 0.110833 \ \mu\text{m}^2, \end{aligned}$$

$$u_c(C) = \sqrt{0.110833} = 0.33292 \ \mu\text{m}.$$

## 8 Step 6: Effective degrees of freedom

The Welch–Satterthwaite formula is

$$\nu_{\text{eff}} = \frac{u_c^4(C)}{\displaystyle\sum_i \frac{\left(c_i u(x_i)\right)^4}{\nu_i}}.$$

The Type B terms have $\nu_i = \infty$ and contribute nothing to the denominator, so only the repeatability term remains:

$$\nu_{\text{eff}} = \frac{u_c^4}{\dfrac{\left(c_{\text{ind}}\, u(C_{\text{ind}})\right)^4}{\nu_A}} = \frac{(0.110833)^2}{\dfrac{(0.005000)^2}{4}} = \frac{0.012284}{6.25 \times 10^{-6}} = 1965.4.$$

## 9 Step 7: Coverage factor and expanded uncertainty

For a 95 % coverage probability, the coverage factor is the two-sided Student-$t$ value $t_{0.975}(\nu_{\text{eff}})$. For $\nu_{\text{eff}} = 1965$ (beyond the $\nu = 1000$ row of the $t$-table) this is

$$k = t_{0.975}(1965) = 1.961 \approx 1.96.$$

The expanded uncertainty is

$$U = k\, u_c(C) = 1.961 \times 0.33292 = 0.6529\ \mu\text{m} \approx 0.65\ \mu\text{m}.$$

With the conventional coverage factor $k = 2$:

$$U = 2 \times 0.33292 = 0.6658\ \mu\text{m} \approx 0.67\ \mu\text{m}.$$

The 95 % coverage interval is

$$[\, C - U,\ C + U \,] = [\, 8.30 - 0.653,\ 8.30 + 0.653 \,] = [\, 7.647,\ 8.953 \,]\ \mu\text{m}.$$

## 10 Uncertainty budget

| Source | Estimate ($\mu$m) | Type | PDF | $u(x_i)$ ($\mu$m) | $c_i$ | $\lvert c_i \rvert u(x_i)$ ($\mu$m) | $\nu_i$ | % of $u_c^2$ |
|---|---|---|---|---|---|---|---|---|
| Repeatability $C_{\text{ind}}$ | 8.30 | A | Normal/$t$ | 0.07071 | 1 | 0.07071 | 4 | 4.51 |
| Calibration $\delta_{\text{cal}}$ | 0 | B | Normal | 0.30000 | 1 | 0.30000 | $\infty$ | 81.20 |
| Resolution $\delta_{\text{res}}$ | 0 | B | Uniform | 0.02887 | 1 | 0.02887 | $\infty$ | 0.75 |
| Alignment $\delta_{\text{align}}$ | 0 | B | Triangular | 0.12247 | 1 | 0.12247 | $\infty$ | 13.53 |
| $C = 8.30\ \mu$m, $u_c = 0.33292\ \mu$m, $\nu_{\text{eff}} = 1965$, $k = 1.96$, $U = 0.65\ \mu$m | | | | | | | | |

The percentage contribution of each input is $(c_i u(x_i))^2 / u_c^2 \times 100$. The calibration uncertainty dominates (81.2 %), followed by the alignment (13.5 %). The repeatability contributes only 4.5 %, and the resolution (0.75 %) is negligible.

## 11 Result

The corrected cylindricity error is

$$\boxed{C = (8.30 \pm 0.65)\ \mu\text{m}}$$

where the number following $\pm$ is the expanded uncertainty $U = k\,u_c$, with combined standard uncertainty $u_c = 0.33\ \mu$m and coverage factor $k = 1.96$ based on $\nu_{\text{eff}} \approx 1965$ effective degrees of freedom, corresponding to a coverage probability of approximately 95 %.